

BLAST FURNACE OPERATIONS MANUAL

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Equipment Description

The blast furnace is used for ironmaking by reducing iron ore using coke and hot air. It is one of the most critical assets in steel production.

Normal Operating Parameters

- Furnace Temperature: 900°C - 1200°C
- Cooling Water Flow: 500 - 800 m³/hr
- Cooling Water Pressure: 95 - 115 bar
- Shell Temperature: Below 350°C

Critical Sensor Thresholds

- Cooling Water Flow < 450 m³/hr
- Shell Temperature > 400°C
- Cooling Pump Vibration > 8 mm/s

Common Failure Modes

1. Tuyere Failure
2. Cooling Water Leakage
3. Cooling Pump Failure
4. Refractory Damage
5. Gas Leakage

Troubleshooting Guide

Low Cooling Water Flow

Possible Causes:

- Pump degradation
- Valve blockage
- Pipeline leakage

Actions:

- Inspect pumps
- Verify valve positions
- Check pressure drop

High Shell Temperature

Possible Causes:

- Cooling failure
- Refractory wear

Actions:

- Increase cooling water circulation
- Inspect cooling circuit

Recommended Maintenance

Daily:

- Verify flow rates
- Check shell temperatures

Weekly:

- Inspect cooling pumps
- Check vibration trends

Monthly:

- Inspect tuyeres
- Review cooling circuit performance